



WINTER TERM MIDTERM I
EXAMINATION
April 16, 2025

DURATION: 3 HOURS

No. Of Students: 22

Department Name & Course Number: Systems and Computer Engineering
SYSC 5108

Course Instructor (s): Changcheng Huang

AUTHORIZED MEMORANDA

Nonprogrammable calculator

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This examination question paper may be taken from the examination room.

**In addition to this question paper, students require: an examination booklet yes
Scantron Sheet no**

Name: _____

Student Number: _____

1. (15%) Given the following dataset. Use the third sample point as the test data and the remaining as training data.

x	0	1	3	6	9
y	-1	2	1	4	2

Find the optimal linear regression model manually. Find the mean squared error for the training data and the mean squared error for test data.

Solution:

$$x = \begin{pmatrix} 1.0 & 1.0 & 1.0 & 1.0 \\ 0 & 1 & 6 & 9 \end{pmatrix}^T$$

$$y = (-1 \quad 2 \quad 4 \quad 2)^T$$

$$(x^T x)^{-1} \approx \begin{pmatrix} 0.55 & -0.07 \\ -0.07 & 0.019 \end{pmatrix}$$

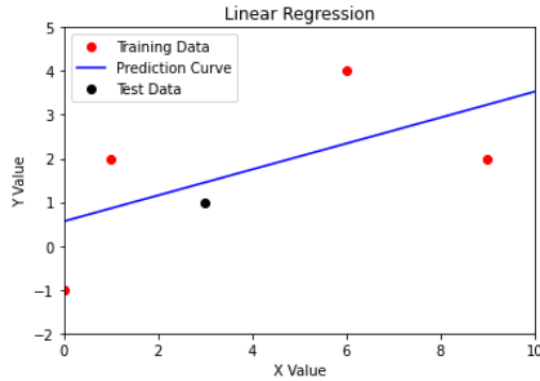
$$\beta \approx (0.56 \quad 0.30)^T$$

$$\hat{y} = (0.56 \quad 0.86 \quad 2.34 \quad 3.23)$$

$$\text{MSE} = 2.0$$

$$\hat{y} = 1.45$$

$$\text{SE} \approx 0.21$$



2. (15%) Consider the requirement to generalize logistic regression from two classes to K classes, i.e., the output $y \in \{1, 2, \dots, K\}$. Assume y follow multinoulli distribution and the logistic regression model can be defined as

$$p(y = k | \mathbf{x}, \beta) = \text{softmax}(z)_k = \frac{\exp(z_k)}{\sum_l \exp(z_l)}$$

where $z_k = \beta_{k0} + \beta_k^T x$, $k = 1, \dots, K-1$ and $z_K = 0$. Then the MLE cost function can be written as

$$J(\beta) = - \sum_{i=1}^M \sum_{k=1}^K I_k(y_i) \log \text{softmax}(z^{(i)})_k$$

Where $I_k(y_i) = 1$, if $y_i = k$, and 0, otherwise. Show that

$$\nabla_{\beta_k} J(\beta_k) = - \sum_{i=1}^M \sum_{k=1}^K I_k(y_i) (1 - \text{softmax}(z^{(i)})_k) \mathbf{x}_i$$

Hint: $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$

Solution:

$$\nabla_{\beta_k} J(\beta_k) = - \sum_{i=1}^M \sum_{k=1}^K I_k(y_i) \nabla_{\beta_k} \log \text{softmax}(z^{(i)})_k$$

In general, we have

$$\nabla_{\beta_k} \log \text{softmax}(z)_k = \frac{\nabla_{\beta_k} \text{softmax}(z)_k}{\text{softmax}(z)_k}$$

where

$$\begin{aligned} \nabla_{\beta_k} \text{softmax}(z)_k &= \nabla_{\beta_k} \frac{\exp(z_k)}{\sum_l \exp(z_l)} = \frac{d}{dz_k} \frac{\exp(z_k)}{\sum_l \exp(z_l)} \nabla_{\beta_k} z_k \\ &= \mathbf{x} \frac{\exp(z_k) (\sum_l \exp(z_l)) - \exp(z_k) \exp(z_k)}{(\sum_l \exp(z_l))^2} = \mathbf{x} \text{softmax}(z)_k (1 - \text{softmax}(z)_k) \end{aligned}$$

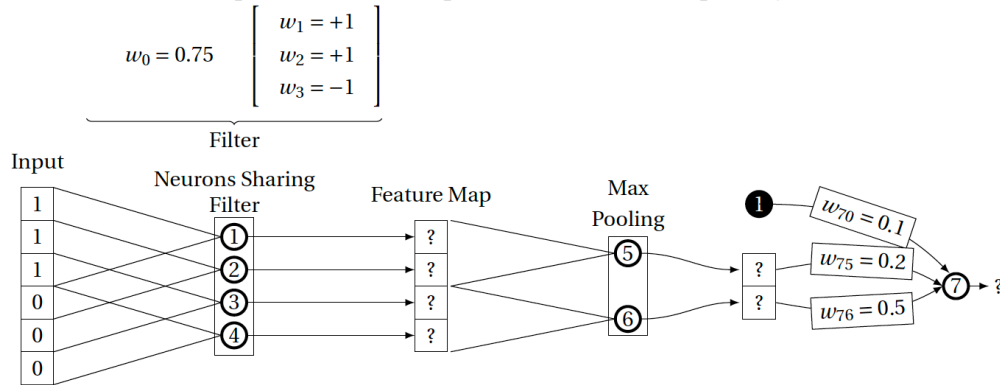
Therefore,

$$\nabla_{\beta_k} \log \text{softmax}(z)_k = \mathbf{x} (1 - \text{softmax}(z)_k)$$

We have

$$\nabla_{\beta_k} J(\beta_k) = - \sum_{i=1}^M \sum_{k=1}^K I_k(y_i) \mathbf{x} (1 - \text{softmax}(z^{(i)})_k)$$

3. (20%) The figure below illustrates a layer of a convolutional neural network that is processing a one-dimensional input. For ease of reference each of the neurons in the network has been labeled: 1; 2; 3; 4; 5; 6; 7. The architecture of the network consists of ReLUs that share a kernel (Neurons 1; 2; 3; 4), followed by a sub-sampling layer containing two max pooling units (Neurons 5; 6), and then a fully connected layer containing a single ReLU (Neuron 7). The kernel in the first layer has a window size 3, and a stride of 1. The max pooling units are 2 to 1, and there is no overlap between the max pooling units.



- (a) Assuming the target output for this input is 2, calculate the δ for each neuron in the network.

Solution:

$$\text{error} = 2 - 1.525 = 0.475$$

Neuron 7

$$\frac{\partial L}{\partial a} = 0.0475 \times -1 = -0.475$$

$$\frac{\partial a}{\partial z} = 1$$

$$\delta_7 = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = -0.475$$

Neuron 6

$$\frac{\partial L}{\partial a} = -0.475 \times 0.5 = -0.2375$$

$$\frac{\partial a}{\partial z} = 1$$

$$\delta_6 = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = -0.2375$$

Neuron 5

$$\frac{\partial L}{\partial a} = -0.475 \times 0.2 = -0.095$$

$$\frac{\partial a}{\partial z} = 1$$

$$\delta_5 = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = -0.095$$

Neuron 4

$$\delta_4 = 0$$

Neuron 3

$$\frac{\partial L}{\partial a} = \delta_6 w_{63} = -0.2375 \times 1 = -0.2375$$

$$\frac{\partial a}{\partial z} = 1$$

$$\delta_3 = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = -0.2375$$

Neuron 2

$$\frac{\partial L}{\partial a} = \delta_5 w_{52} = -0.095 \times 1 = -0.095$$

$$\frac{\partial a}{\partial z} = 1$$

$$\delta_2 = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = -0.095$$

Neuron 1

$$\delta_1 = 0$$

(b) Calculate the weight update for w_{75} and w_1

Solution:

$$\frac{\partial L}{\partial w_{75}} = \delta_7 a_5 = -0.475 \times 2.75 = -1.31$$

$$\frac{\partial L}{\partial w_1} = 0 - 0.095 - 0.2375 + 0 = -0.3325$$

4. (10%) Suppose $\gamma = 0.9$ and the reward sequence is $R_1 = 2$ followed by an infinite sequence of 7s. What are G_0 and G_1 ?

Solution: $\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$, we get $G_1 = 70$. $G_0 = 2 + 0.9 \cdot G_1 = 65$.

5. (20%) Consider a two-state-two-action reinforcement learning scenario with the following environment.

State	Action	Next State	Probability	Reward
0	0	0	1	-1
0	1	1	1	1
1	0	1	1	1
1	1	0	1	-1

Assume $\gamma = 0.1$ and an ϵ -greedy policy with $\pi(1|0) = \pi(0|1) = 0.6$.

- (a) Starting from state 0 and 1 at time 0 respectively, generate two trajectories using the following random number sequence and terminate at $t = 3$. Assume $\alpha = 0.6$, estimate the state-value function and action-value function with Monte-Carlo first-visit method using the generated trajectories. Indicate clearly which random numbers you have used.

0.94 0.27 0.30 0.18 0.75 0.35

Solution:

Starting from state 0, RN $0.94 > 0.4$, action 1, next state 1, RN $0.27 < 0.6$, action 0, next state 1, RN $0.30 < 0.6$, action 0, next state 1. Trajectory $\tau_0 = 0111$, rewards 111

Starting from state 1, RN $0.18 < 0.6$, action 0, next state 1, RN $0.75 > 0.6$, action 1, next state 0, RN $0.35 < 0.4$, action 0, next state 0. Trajectory $\tau_1 = 1100$, rewards 1-1-1

$$v_{\pi}(0) = 1 + 0.1 \times 1 + 0.01 \times 1 + 0.6 \times [-1 - 1 - 0.1 \times 1 - 0.01 \times 1] = -0.156$$

$$v_{\pi}(1) = 1 + 0.1 \times 1 + 0.6 \times [1 + 0.1 \times (-1) + 0.01 \times (-1) - 1 - 0.1 \times 1] = 0.974$$

Using similar logic to state-value function, we have

$$q_{\pi}(s, a) = q_{\pi}(s, a) + \alpha[G_t - q_{\pi}(s, a)]$$

We have

$$q_{\pi}(0,0) = -1$$

$$q_{\pi}(0,1) = 1 + 0.1 \times 1 + 0.01 \times 1 = 1.11$$

$$q_{\pi}(1,0) = 1 + 0.1 \times 1 + 0.6 \times [1 + 0.1 \times (-1) + 0.01 \times (-1) - 1 - 0.1 \times 1] = 0.974$$

$$q_{\pi}(1,1) = -1 + 0.1 \times (-1) = -1.1$$

- (b) Assume $\alpha = 0.6$, starting with all action values equal to 0, find action values using off-policy TD control with the trajectories generated in (a)

Solution:

Following the off-policy TD control:

$$q(s_t, a_t) = q(s, a) + \alpha[r + \gamma \max_{a'} q(s', a') - q(s, a)]$$

We have:

Trajectory τ_0

$t = 0$

$$q(0,1) = 0 + 0.6 \times 1 = 0.6$$

$t = 1$

$$q(1,0) = 0 + 0.6 \times [1 + 0.1 \times 0 - 0] = 0.6$$

$t = 2$

$$q(1,0) = 0.6 + 0.6 \times [1 + 0.1 \times 0.6 - 0.6] = 0.876$$

Trajectory τ_1

$t = 0$

$$q(1,0) = 0.876 + 0.6 \times [1 + 0.1 \times 0.876 - 0.876] = 1.003$$

$t = 1$

$$q(1,1) = 0 + 0.6 \times [-1 + 0.1 \times 0.6 - 0] = -0.564$$

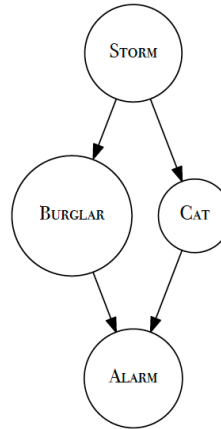
$t = 2$

$$q(0,0) = 0 + 0.6 \times [-1 + 0.1 \times 0.6 - 0] = -0.564$$

6. (20%) Stormy nights are rare. Burglary is also rare, and if it is a stormy night, burglars are likely to stay at home (burglars don't like going out in storms). Cats don't like storms either, and if there is a storm, they like to go inside. The alarm on your house is designed to be triggered if a burglar breaks into your house, but sometimes it can be set off by your cat coming into the house, and sometimes it might not be triggered even if a burglar breaks in (it could be faulty or the burglar might be very good).

- (a) Model the relationship between stormy night or not, burglars at your house or not, cats at your house or not, alarm triggered or not as a directed model and show the directed graph.

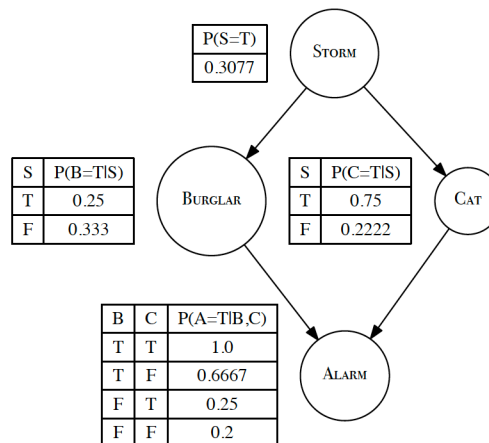
Solution: The figure below illustrates a Bayesian network that encodes the describe causal relationships. Storms directly affect the behavior of burglars and cats, and this is reflected by links from the storm node to the burglar and cat nodes. The behavior of burglars and cats both affect whether the alarm goes off, and hence there are links from each of these nodes to the alarm node.



- (b) Using the following examples to estimate the probabilities that characterize the directed graph. Show the probabilities in the graph.

ID	STORM	BURGLAR	CAT	ALARM
1	false	false	false	false
2	false	false	false	false
3	false	false	false	false
4	false	false	false	false
5	false	false	false	true
6	false	false	true	false
7	false	true	false	false
8	false	true	false	true
9	false	true	true	true
10	true	false	true	true
11	true	false	true	false
12	true	false	true	false
13	true	true	false	true

Solution:



- (c) What value will the Bayesian network predict for ALARM, given that there is both a burglar and a cat in the house but there is no storm?

Solution: Because both the parent nodes for ALARM are known, the probability distribution over ALARM is independent of the feature STORM. Consequently, we can read the relevant probability distribution over ALARM directly from the conditional probability table for the ALARM node.

Examining the conditional probability table, we can see that when BURGLAR = true, and CAT = true, then ALARM = true is the MAP prediction. In other words, the network would predict that the alarm would sound in this situation.

- (d) Generate a sample that includes a random outcome for each random variable in the network by following the convention that a positive outcome is the state 1 while the negative is state 0 (e.g., for random variable STORM: true = 1, false = 0) with the following random number sequence. Indicate clearly which random numbers you have used.

0.94 0.27 0.30 0.18

Solution:

$0.94 > 1 - 0.3077$, we have $S = T$

$0.27 < 1 - 0.25$, we have $B = F$

$0.30 > 1 - 0.75$, we have $C = T$

$0.18 < 1 - 0.6667$, we have $A = F$

The final result will be

$STORM = T, BURGALAR = F, CAT = T, ALARM = F.$